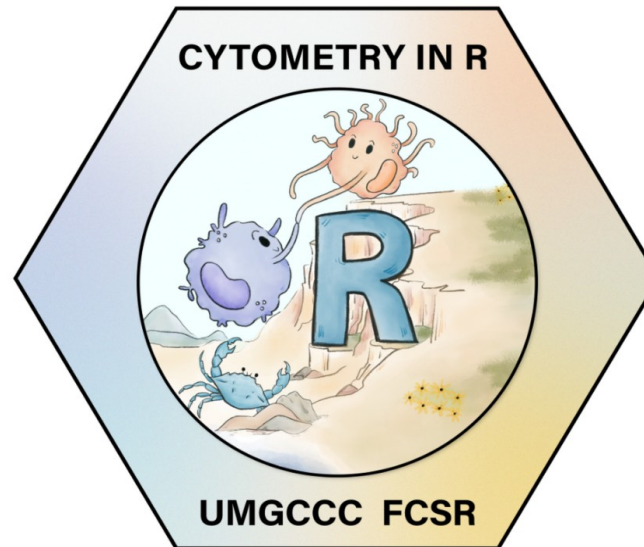




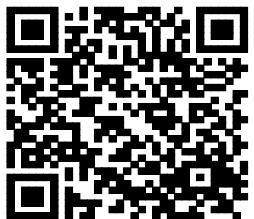
Cytometry in R: A free weekly course for flow cytometrist with no-to-little coding experience



David Rach, Natarajan Ayithan, Xiaoxuan Fan.

UMGCCC Flow Cytometry Shared Resource

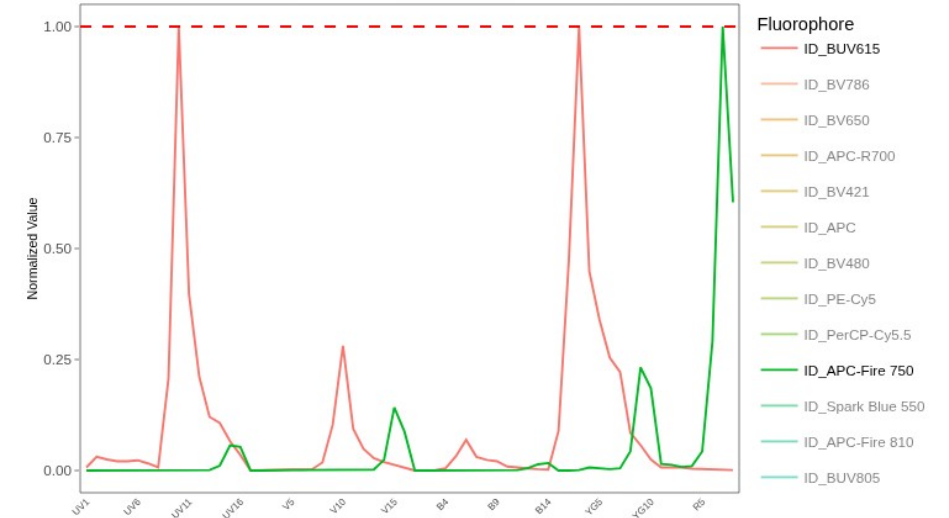
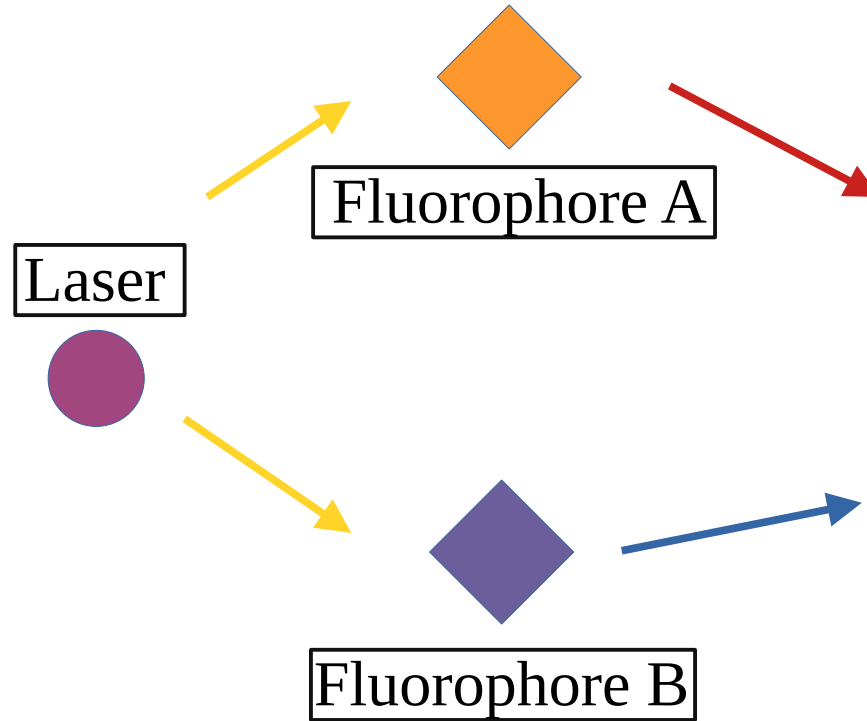
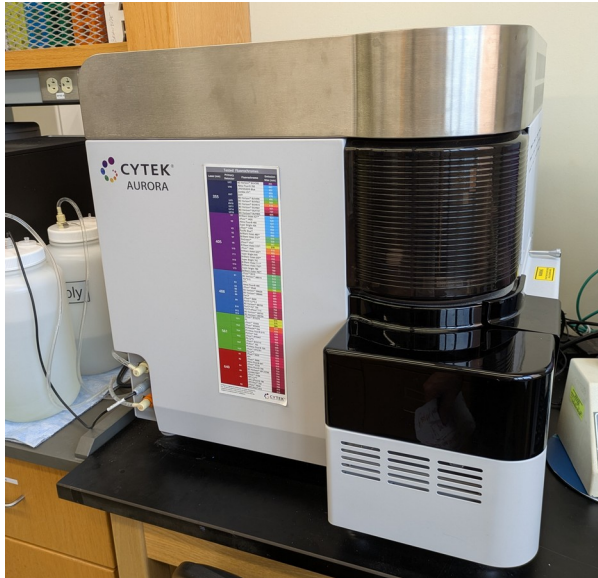
August 10, 2026



Website

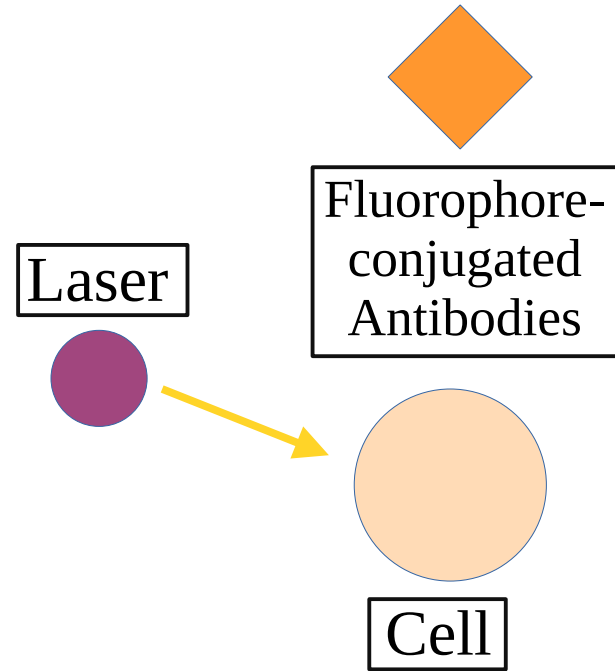
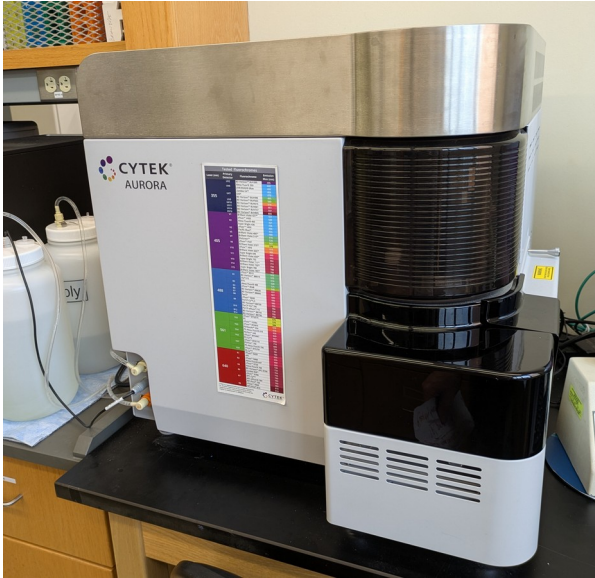


Flow Cytometry 101



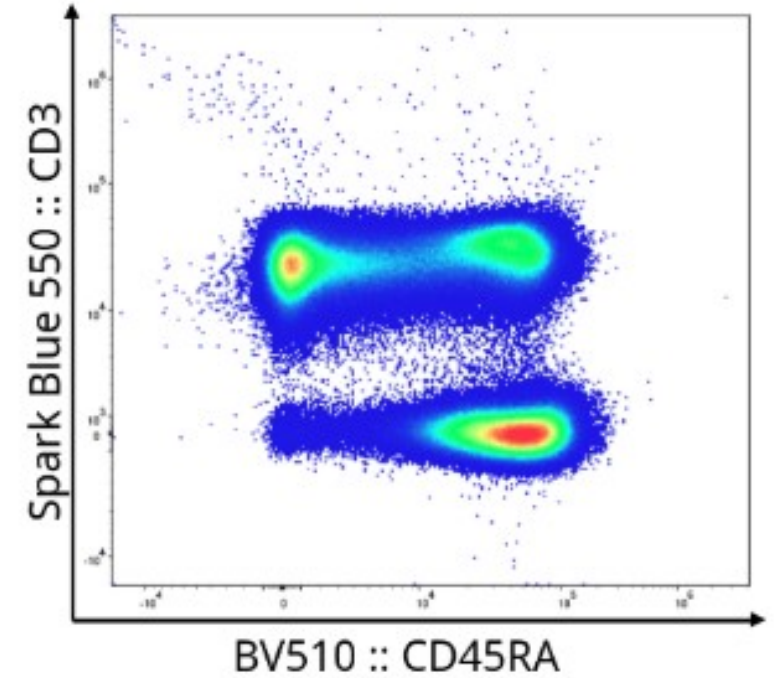
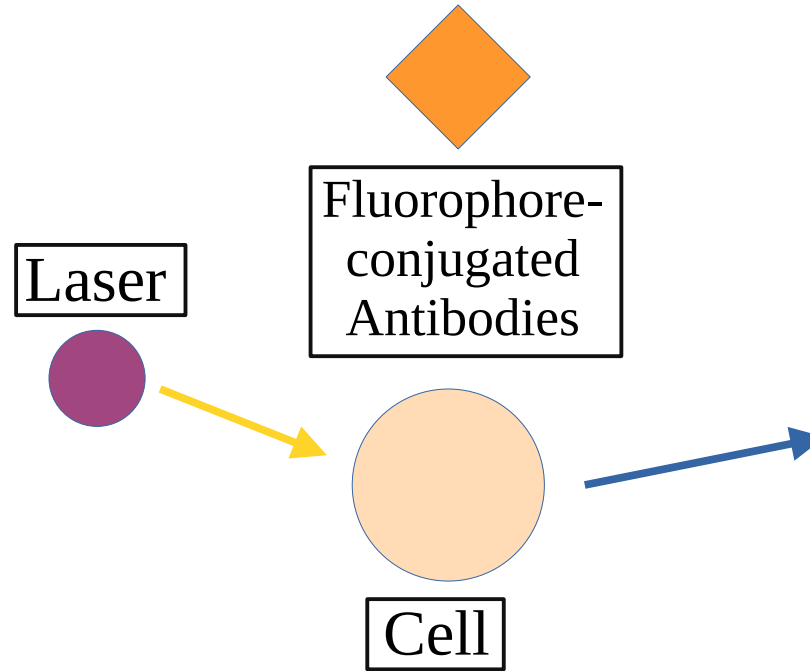
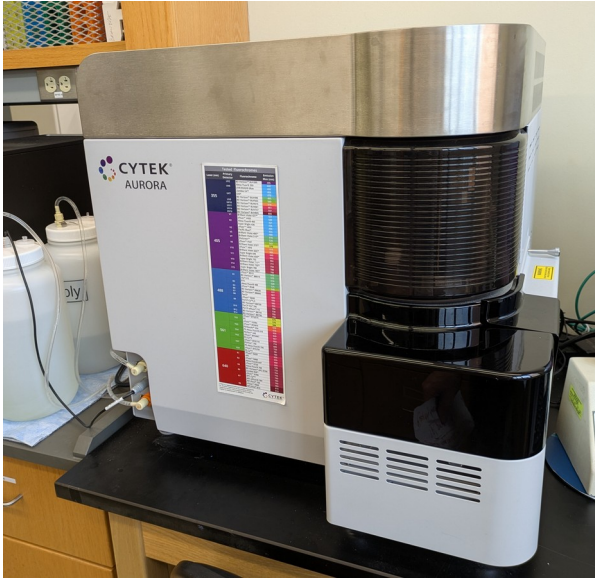
Shine a laser on a fluorophore, it emits back light in a distinct pattern

Flow Cytometry 101



- ❑ Fluorophores are conjugated to monoclonal antibodies.
- ❑ When fluorophore-antibody mix is added to cells in suspension, antibodies bind to cells that express their antigen.

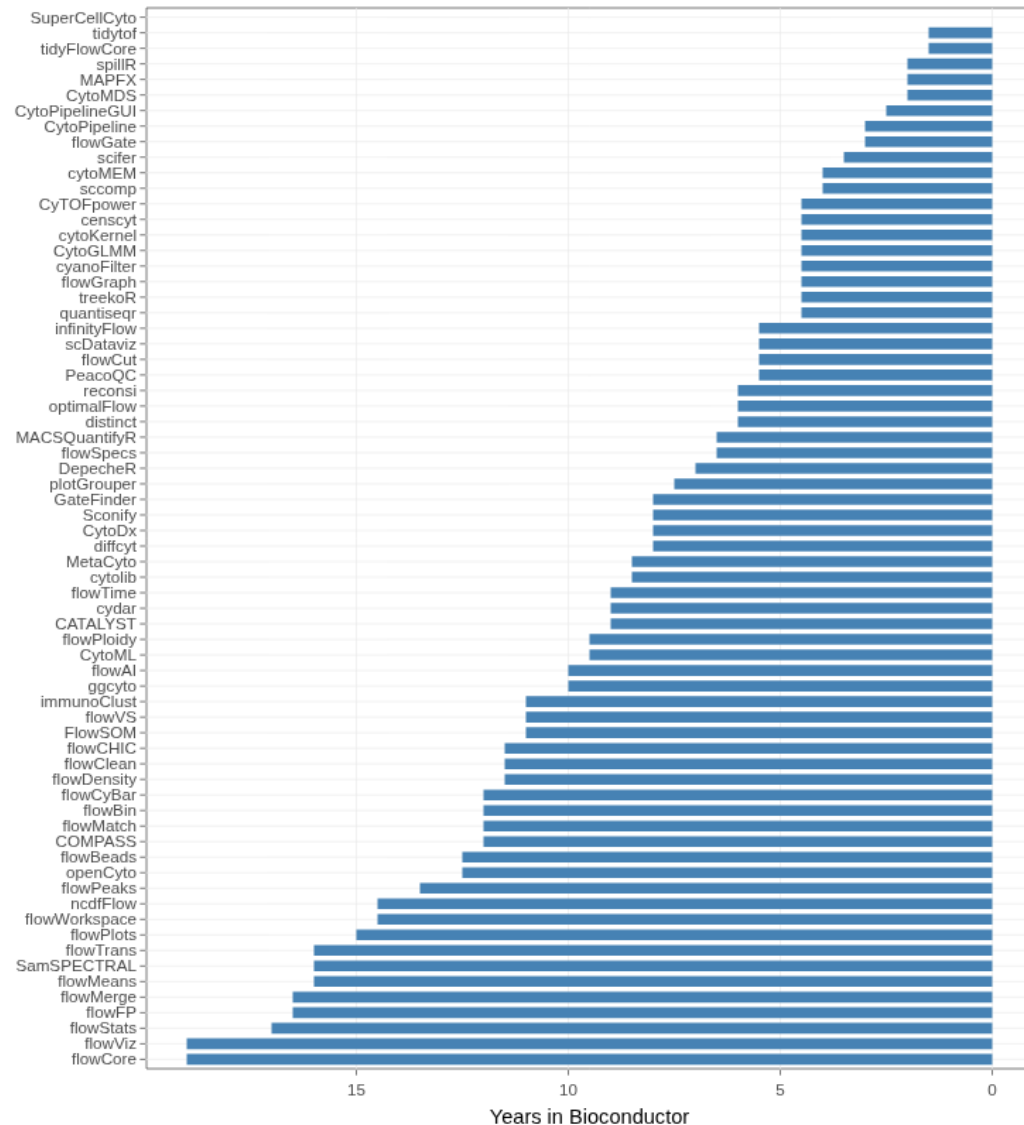
Flow Cytometry 101



- Shine a laser on these now stained cells, the fluorophores-conjugated to the antibodies light up, and we can infer marker expression

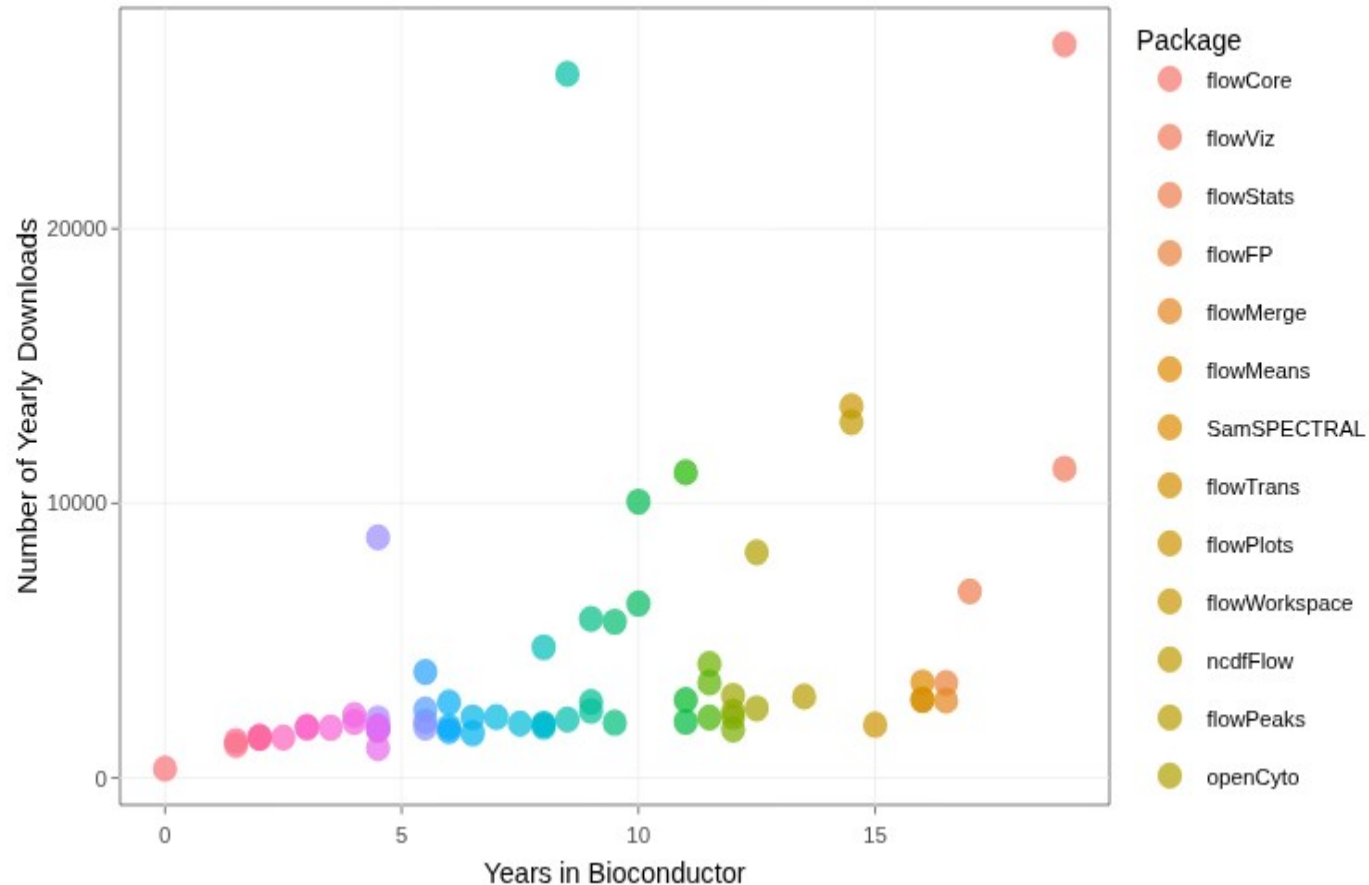
Flow Cytometry and Bioconductor

Bioconductor Flow Cytometry R packages



- FlowCore is the oldest extant package (Bioc 2.0, 2007)
- 72 packages list Flow Cytometry in their biocViews

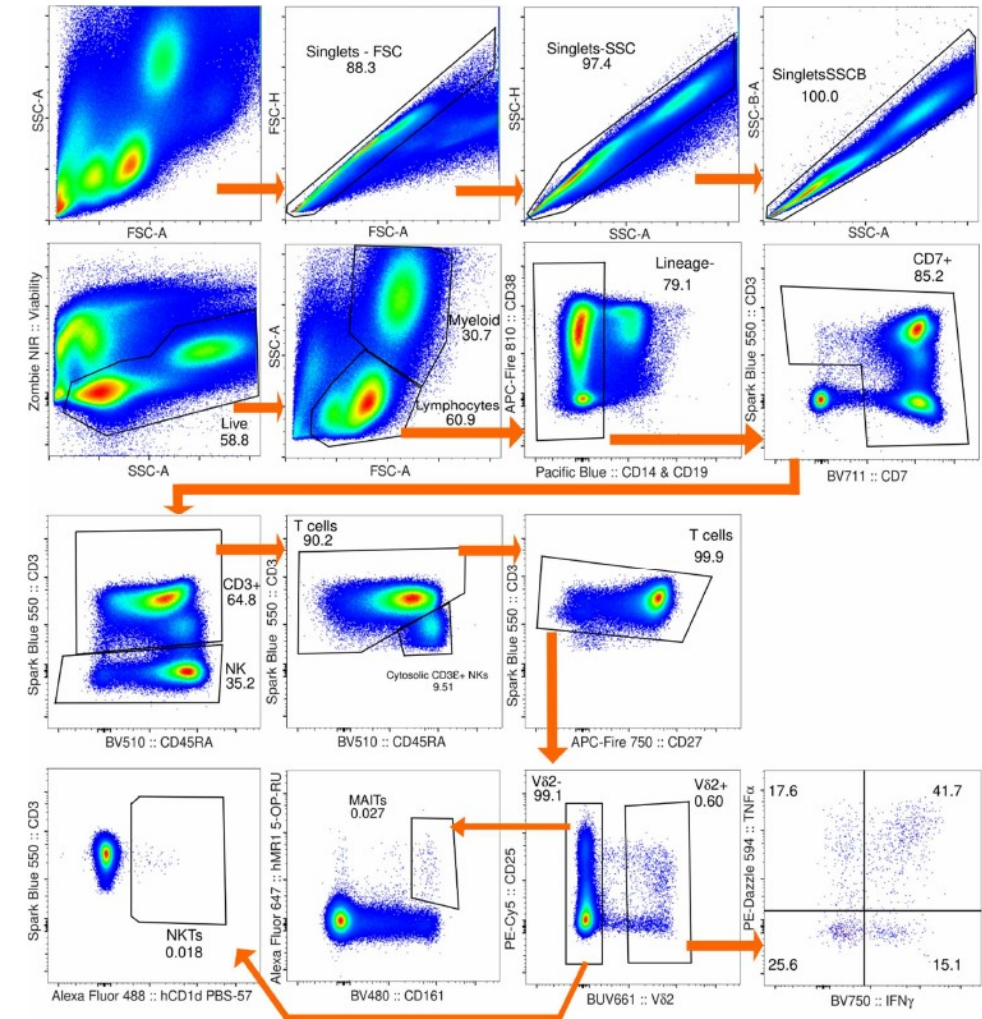
Flow Cytometry and Bioconductor



- Most are severely under-utilized
- 7 packages with $> 10,000$ yearly downloads
- 15 packages with > 4000 yearly downloads

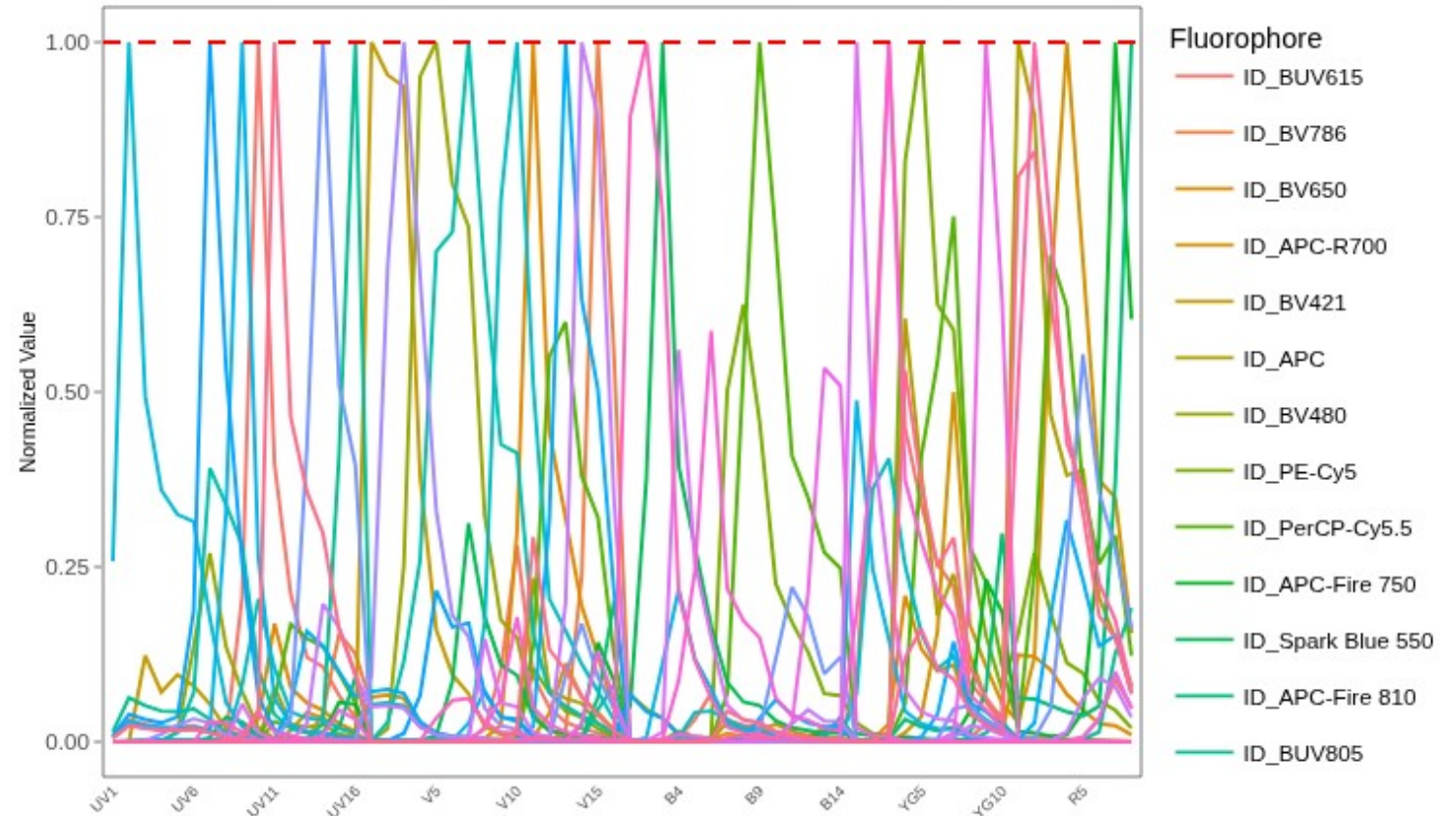
Lack of wider community adoption

- Vast majority of the community uses commercial software (ex. FlowJo, FCSExpress, etc.)
- Manual hierarchical gates are drawn around cell populations of interest via graphical-user interface



Spectral Flow Cytometry

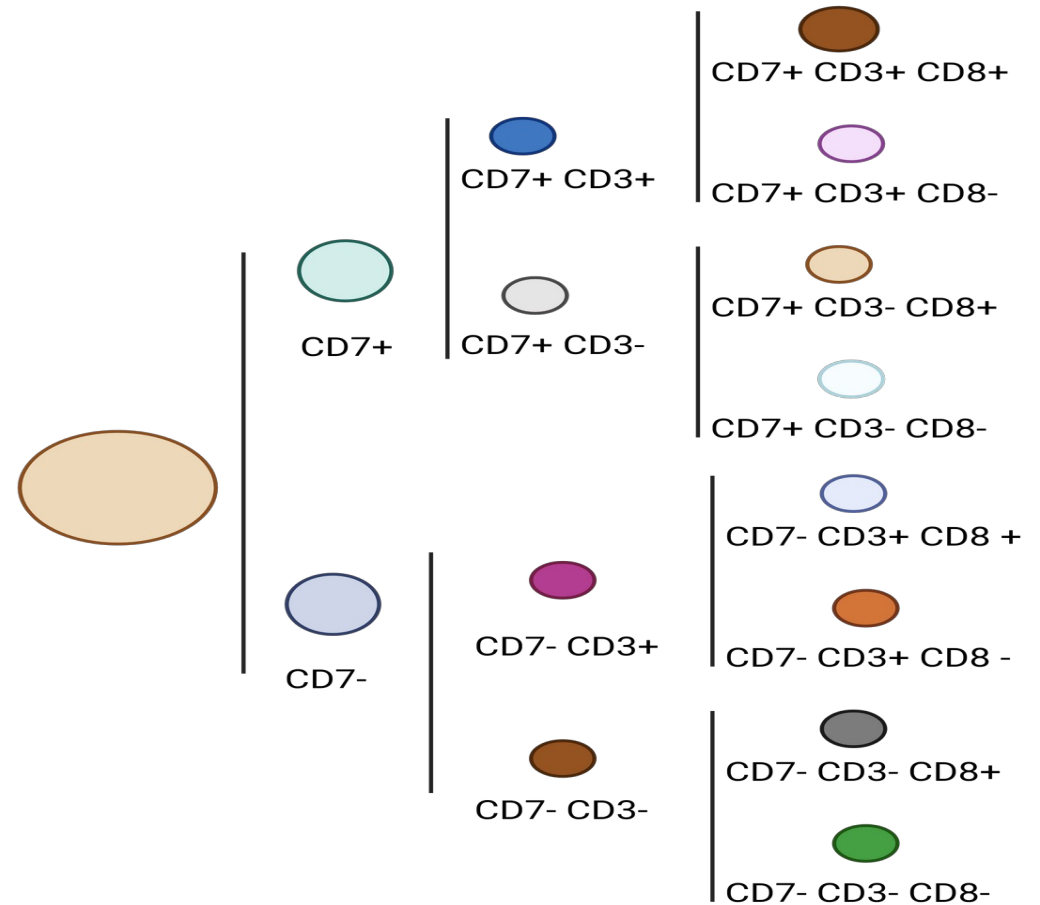
- Popularized by the Cytex Aurora
- Able to resolve incredibly similar fluorophores
- Largest panels >60 fluorophores, matching mass cytometry
- Retains speed of acquisition of conventional flow cytometry



Complex Datasets, Combinatorially Exponential Clusters

Manual Gating no longer is sufficient to thoroughly profile individual specimens

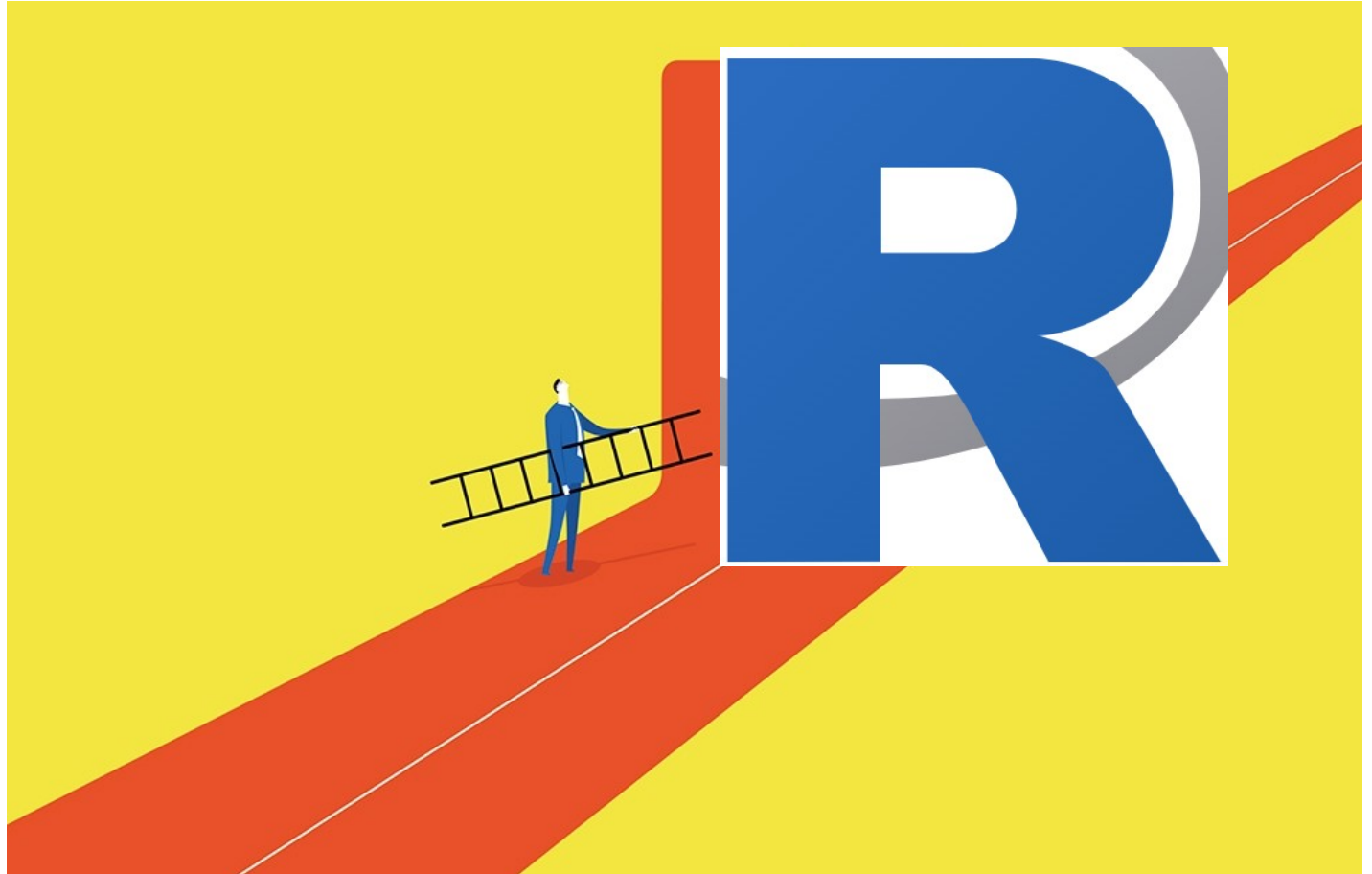
- If every marker split positive and negative; splits repeated for every marker:
 - Potential groupings: $2^{\text{NumberOfFluorophores}}$
- **9-color** CFC panel: **512** groups at most
- **60-color** SFC panel: **1.2e+18** clusters at most (**quintillion**)



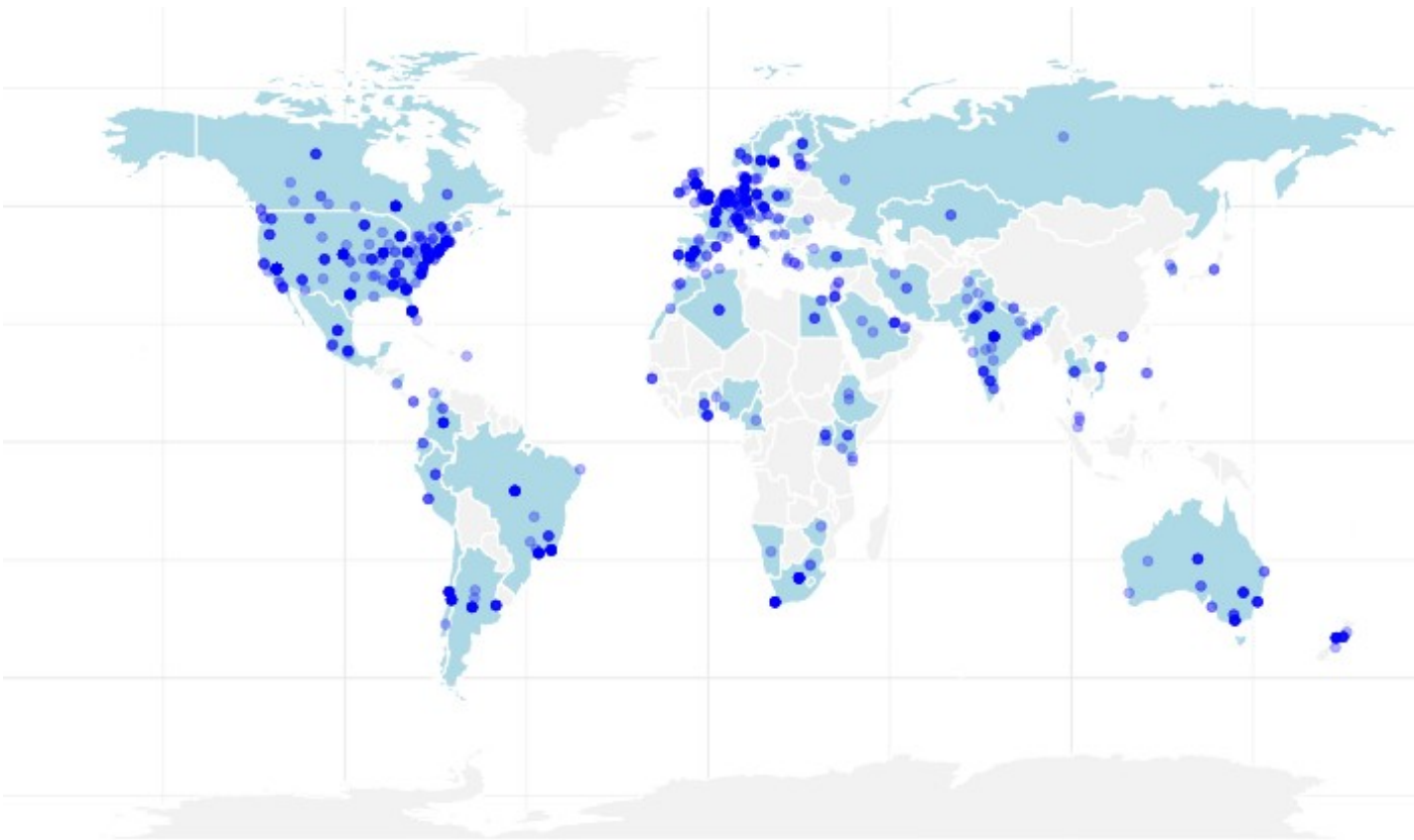
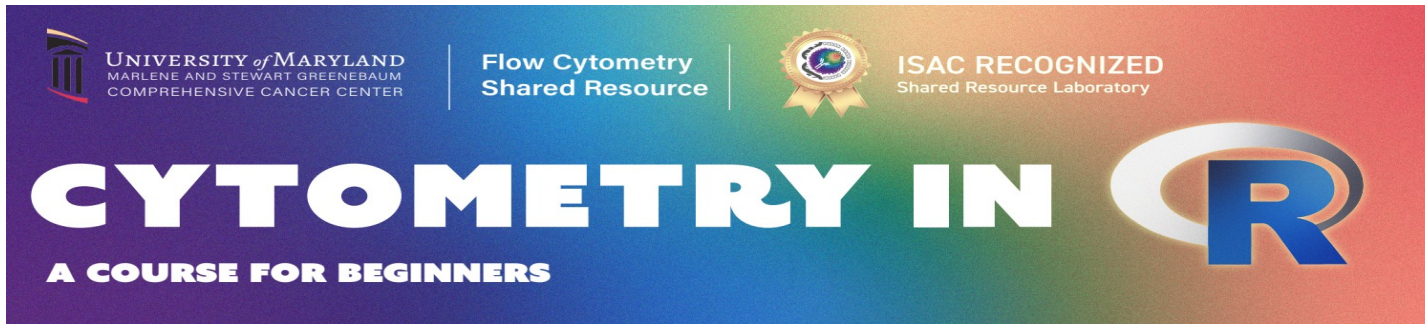
Increasing use of semi-supervised and unsupervised algorithms for analysis, many already implemented as Bioconductor packages

Major barrier to entry

- Few resources to begin with
- Those that exist aimed at those with intermediate-advanced R skills, not beginners
- Know-how concentrated at a few institutions
- Most self-learning attempts end in frustration



Cytometry in R: A free weekly course for flow cytometrist with no-to-little coding experience



- Announced in November 2025
- Woke up to find it went viral
- 2000 researchers worldwide signed up for email updates
- Had to update planned format

Overall structure



- ❑ 30-week course, starting from Intro to R, gradually moving to more advanced content.
- ❑ Multiple ways to learn with flexible scheduling
- ❑ Taught in Positron instead of Rstudio (ask me why!)
- ❑ Version Control/Git introduced from the beginning

YouTube

UMGCCC FCSR
@CytometryInR • 848 subscribers • 21 videos
University of Maryland Greenebaum Comprehensive Cancer Center Flow Cytometry Shared Resource
umgcccfcscr.github.io/CytometryInR
Subscribed

Live Playlists Posts

Latest Popular Oldest

PhD Dissertation Defense - David Rach
Scheduled for 6/17/26, 12:00 PM
Notify me

11 - Data for Statistics
Cytometry Core • 11 - Data for Statistics
AUTHOR David Rach
PUBLISHED May 5, 2026
101 views • Streamed 3 weeks ago

10 - Downsampling and Concatenation
Cytometry Core • 10 - Downsampling and Concatenation
AUTHOR David Rach
PUBLISHED April 27, 2026
129 views • Streamed 1 month ago

09 - It's Raining Functions!
Cytometry Core • 09 - It's Raining Functions!
AUTHOR David Rach
PUBLISHED April 13, 2026
172 views • Streamed 1 month ago

08 - Manual and Automated Gating
Cytometry Core • 08 - Manual and Automated Gating
AUTHOR David Rach
PUBLISHED April 7, 2026
226 views • Streamed 1 month ago

07 - Applying Transformations
Cytometry Core • 07 - Applying Transformations
AUTHOR David Rach
PUBLISHED March 26, 2026
163 views • Streamed 2 months ago

06 - Visualizing with ggplot2
Cytometry Core • 06 - Visualizing with ggplot2
AUTHOR David Rach
PUBLISHED March 17, 2026
228 views • Streamed 2 months ago

05 - Gating Sets
Cytometry Core • 05 - Gating Sets
AUTHOR David Rach
PUBLISHED March 1, 2026
253 views • Streamed 2 months ago

- Three live-streams a week (accommodating different timezones and schedules)
- Recordings available immediately after
- Best of the weekly livestreams added to the playlist, cutting down on editing



Website

- ❑ Created with Quarto, hosted via GitHub pages.
- ❑ Code, code outputs, screenshots and explanations
- ❑ Searchable
- ❑ New weekly material added as a new .qmd file

The screenshot displays the 'Cytometry i...' website. The top navigation bar includes links for 'About', 'Schedule', 'Course', 'Announcements', 'Existing Resources', 'Discussions', and 'Homeworks'. A light blue banner at the top states: 'See Course tab for Course Materials, see YouTube tab for livestream recordings'. The main content area is divided into three sections:

- Getting Started**: A sidebar menu with links to 'Intro to R', '01 - Installing R Packages', '02 - File Paths', '03 - Inside a .FCS file', '04 - Intro to Tidyverse', '05 - Gating Sets', '06 - Visualizing With ggplot2', '07 - Applying Transformations', '08 - Ways to Gate' (highlighted), and '09 - It's Raining Functions!'. Below this are links for 'Cytometry Core', 'Beyond the Sandbox', 'The World is your Oyster', 'Bonus Content', and 'Community'.
- Code**: A central panel showing R code to create a GatingSet object. The code includes:

```
SFC_cytoiset <- load_cytoiset_from_fcs(fcs_files, truncate_max_range = FALSE, transformation = FALSE) SFC_GatingSet <- GatingSet(SFC_cytoiset) SFC_Parameters <- colnames(SFC_GatingSet) FluorophoresOnly <- SFC_Parameters[!stringr::str_detect(SFC_Parameters, "FLUOROPHORE")] Biexponential <- flowjo_biexp_trans(channelRange=4096, maxValue=26000, pos=4.5, neg=2, widthBasis=-500) MyBiexTransform <- transformerList(FluorophoresOnly, Biexponential) transform(SFC_GatingSet, MyBiexTransform)
```

 Below the code, it says 'A GatingSet with 6 samples' and 'And now, let's apply the GatingTable scheme. As we create and click done, the next gate will automatically pop up in the Shiny App.' The code

```
gs_apply_gating_strategy(SFC_GatingSet, gating_strategy = GatingTable)
```

 is also shown.
- On this page**: A sidebar menu with links to 'Background', 'Walk Through', 'Setup', 'Manual - flowGate', 'Polygon', 'Rectangle', 'Span', 'Quadrant', 'Redraw Gate', 'Saving Gates', 'Rapid Gate Deployment.' (highlighted), 'Visualizing Gates', 'Summary', 'Automated - openCyto', 'Take Away', 'Additional Resources', and 'Take-home Problems'.

At the bottom, there is a 'Draw your gate' section with a 'Reset' button, a 'Gate Type' dropdown (set to 'Polygon'), and input fields for 'X Minimum', 'X Maximum', 'Y Minimum', and 'Y Maximum'. To the right is a flow cytometry plot showing a population of cells with a gate drawn around them. The plot is titled 'Gate Name: singlets' and 'subset of: root'.

GitHub

The screenshot shows the GitHub repository page for 'CytometryInR' by 'UMGCCCCFSR'. The repository is public and has 11 watchers, 603 forks, and 388 stars. The main branch is 'main'. The repository contains a file tree with folders like 'course', 'docs', and 'images', and files like '.gitignore', 'Announcements.qmd', 'CODE_OF_CONDUCT.md', 'CONTRIBUTING.md', 'ExistingResources.qmd', 'Homeworks.qmd', 'LICENSE', 'LICENSE_CC-BY-SA', 'PackageWalkthroughs.qmd', 'README.md', 'Schedule.qmd', '_quarto.yml', and 'index.qmd'. The 'About' section includes a 'Course Website' link, a 'Readme', and a 'Code of conduct'. The 'Contributors' section lists 'DavidRach' and 'UMGCCCCFSR'. The 'Languages' section shows a bar chart with R (85.4%), SCSS (12.4%), JavaScript (1.5%), and GLSL (0.7%).

File/Folder	Description	Last Commit
course	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
docs	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
images	Community Package Walkthrough for TRU-OLS	last week
.gitignore	Start of course material transfer	5 months ago
Announcements.qmd	Week 11 slides	last month
CODE_OF_CONDUCT.md	Create CODE_OF_CONDUCT.md	5 months ago
CONTRIBUTING.md	Create CONTRIBUTING.md	5 months ago
ExistingResources.qmd	Existing Resources Links	2 months ago
Homeworks.qmd	Week 11 slides	last month
LICENSE	Start of course material transfer	5 months ago
LICENSE_CC-BY-SA	Start of course material transfer	5 months ago
PackageWalkthroughs.qmd	Updated link to Week #3 recording	4 months ago
README.md	Downsampling - Week 10. Slides and Concatena...	2 months ago
Schedule.qmd	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
_quarto.yml	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
index.qmd	Updated in-person schedule, removed course st...	4 months ago

- ❑ Code and datasets available via the GitHub repository
- ❑ Course participants start course by creating GitHub account and forking the repository
- ❑ After Positron is linked to GitHub account, pull locally
- ❑ When new material posted each week, sync and pull



Discussion Forum

- GitHub Discussions page has been repurposed as a community forum for questions, etc.

The screenshot displays the GitHub Discussions interface for the UMGCCCFCSR repository. At the top, navigation links include Discussions, Actions, Security and quality, Insights, and Settings. Below this, four category tiles are visible: Introductions, Week # 11 - General Discussion Thread, Cytometry in R - Week # 11 - Data for Statistics, and Cytometry in R - Course Update and Bonus... Each tile shows its respective icon and the repository name UMGCCCFCSR.

Below the tiles is a search bar with the text "is:open" and a "New discussion" button. To the left of the main discussion list is a "Categories" sidebar with options: View all discussions, Announcements, General, Ideas, Introductions, Polls, Q&A, and Show and tell. Below these is a "Most helpful" section listing a discussion by mpatgaonkar-coder.

The main "Discussions" list on the right contains several entries, each with a lock icon, a title, a user profile, a timestamp, and a comment count. The discussions include:

- Cytometry in R - Week # 07 - Applying Transformations (Week07) by UMGCCCFCSR, announced on Mar 24 in Announcements, 0 comments.
- Questions about tidy data by alexandragabrielaferreira, asked on Mar 24 in Q&A, Unanswered, 0 comments.
- Pull error on Week 06 branch by denisebritoh, started on Mar 23 in General, 0 comments.
- Copying index file by Oscar-Kai, asked on Mar 23 in Q&A, Unanswered, 0 comments.
- Week #1 - General Discussion Thread (Week01) by DavidRach, started on Feb 5 in General, 26 comments.
- Cytometry in R - Week # 6 - Visualizing with ggplot2 (Week 06) by UMGCCCFCSR, announced on Mar 17 in Announcements, 0 comments.
- GitHub personal access token (Week0) by DavidRach, asked on Jan 23 in Q&A, Answered, 50 comments.
- clock appears when trying to pull from git & commit by MjApo123, started on Mar 6 in General, 3 comments.
- Bioconductor Cytometry R Packages (Week01) by DavidRach, started on Feb 8 in Show and tell, 3 comments.
- No Class This Week by UMGCCCFCSR, announced on Mar 10 in Announcements, 0 comments.

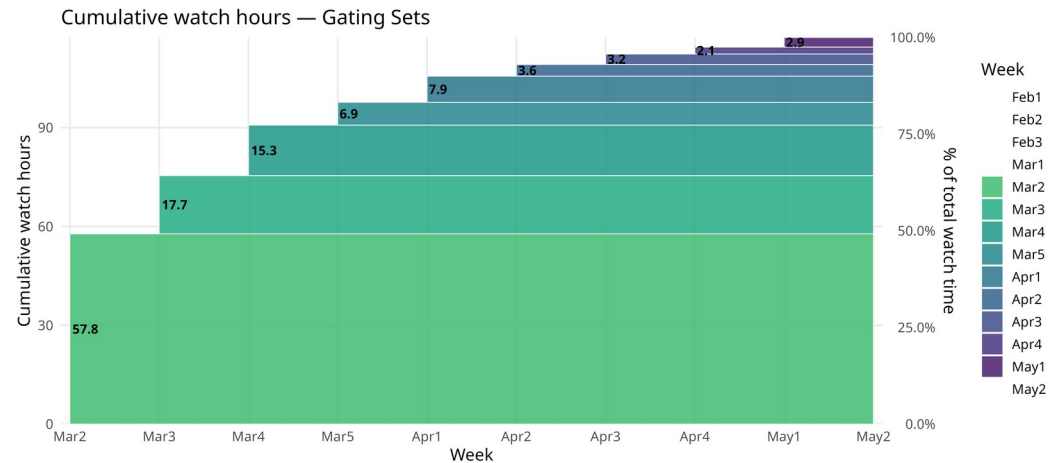
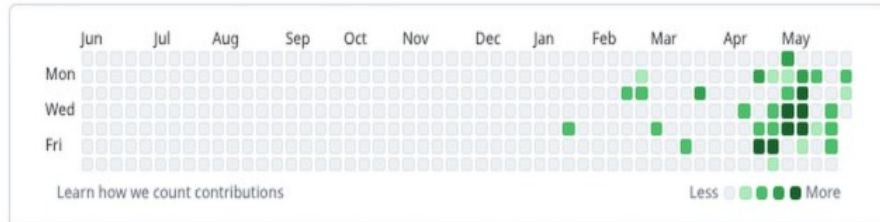
At the bottom left, there is a link to "ins/categories/announcements".



How has it been going?

143 contributions in the last year

2026



□ Per Google, GitHub and YouTube analytics:

- 897 Subscribers
- 638 Forks
- 200-350 average weekly participants
- Spread out at own pace

Community Feedback Surveys: What has worked well?

Biggest thing is thank you so much for providing this resource for free! It's a precious opportunity to learn something that I really hope is something I can build on and use to my advantage after my PhD when seeking job opportunities!



You explain the logic and you start from "zoom out" and only then go into "zoom in", which helps a lot for a biologist to stay in focus and stay interested. I liked that a lot. You also explain very well the pros and cons of each strategy. Honestly, I can't find time to practice but I do listen to all your youtube channel recordings and hope to practice in future.



The course adapts to every modality of learning, by reading, watching, re-visiting documentation, online, in person. So great organization. A lot of work for organisers, thank a lot



Going through the "boring" R fundamentals in a methodical way before getting to the hardcore cytometry content has been really helpful



Amazing work preparing the course material! It is really helpful to have all those sources of information and different options to access the material



Everything! This course is a great opportunity to learn. It is a very generous thing, too, because is completely free with a lot of work behind, so many many thanks for the amazing work.



Each class does a very good job of going through things step by step, and fully explaining what (and why) the code is doing what it does



Everything is so well explained. With details and good walkthrough. I like to have the print screens to see what we need to follow. It must have been so much work. I also like that we can learn at our pace. Working in a lab doesn't always allow us to have time to do thoses kind of stuff



Being able to use real data and reproduce the same results and graphs on my own computer. Let's see when i try with my own, I think I will be capable, fingers crossed.



The structure of the learning has made sense to me so far. I like the way the content is available in multiple formats, which allows me to use whichever format I feel may be most helpful that week for the content. I think the layout on the website is very helpful and easy to navigate.



The website is great, everything is clear, detailed, and easy to find and navigate. David is great at teaching and I really like how reassuring he is with how hard this content can be for some of us. Offering the course for free is also very generous.



The lessons have been clear with good pacing. Take home problems are challenging but not impossible. I also like the breaks to catch up and delve deeper in certain subjects.



- Flexible schedule, with
- multiple formats
- Detailed explanations from a cytometry perspective
- Strangely... Positron and Git?
- Free

Community Feedback Surveys: What can be improved on?

- In hindsight: Probably shouldn't have started this project at the same time as writing my thesis.
- Case-in-point: Feedback on the Optional Take-Home Problems
- Despite planning for it, still vastly underestimated the number of potential installation issues on Windows vs. individual IT policies

The homework problems? I imagine that this is a huge amount of extra work and that's why there is no a feedback. However, the course is pretty well organized and easy to follow up.



Following along with the livestream can be a bit tricky, some sessions have felt a bit too content-heavy for the 1hr time slot. Also, as we jump from topic to topic each week it can be hard to consolidate some of that learning!



Since it's the first time doing this curse it's difficult to anticipate problems that might need troubleshooting, but maybe adding a few more "frequent problems" section or something of the sort for future instances of the program.



Instead of screenshots, I would prefer a live stream of your monitor so I know exactly what you are selecting. I am constantly referring to old notes to figure out if I'm pulling/ pushing/ forking data and I'm still not sure if I'm doing this correctly.



As completely new to R/ coding, I need more time understanding the functions within the packages. in short, I need time to play with code functions. Contasting to my own problem, the course stretched a bit long, since peoples profession and life changes, we may miss some parts of the course.



Answering questions faster in the Discussion board if it is holding someone up from moving forward with the course would be helpful, as well as getting input on submitted homework to know if we're on the right track, or if there is something we missed, or could get to the answer in a better way.



Where to next?

Schedule

Introduction to R



Installing R Packages



File Paths



Inside an .FCS file



Intro to Tidyverse



Gating Sets



Visualizing with ggplot2



Applying Transformations



Manual & Automated Gating



It's Raining Functions!



Downsampling & Concatenation



Data for Statistics



Spectral Signatures



Similarities & Hotspots



Unmixing in R



Cleaning Algorithms

Cytometry Core



Unsupervised Analysis



Clustering Algorithms



Normalization Algorithms



Dimensionality Visualization



Annotating Unsupervised Clusters



The Art of GitHub Diving



XML Files All The Way Down



Utilizing Bioconductor Packages



Building your First R Package



Everyone gets a Quarto Website



Reproducibility & Replicability



Open Source Licenses



Validating Algorithmic Tools



Databases & Repositories



Assembling Web Data

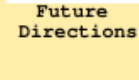


Future Directions

Beyond the Sandbox

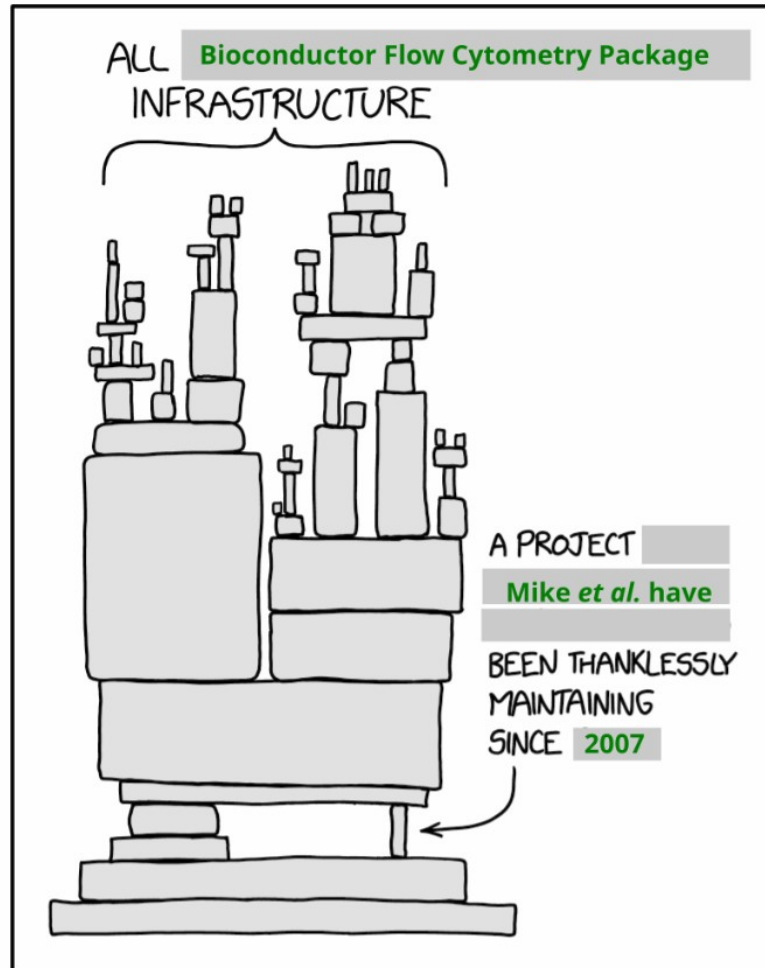


The World's your Oyster



- ❑ Wrapping up Cytometry Core module
- ❑ Starting on Unsupervised Analysis section

A rising tide floats all boats



- ❑ 72 Bioconductor cytometry packages (+ 20? GitHub based packages)
- ❑ Only subset will have been covered during the planned 30 week course
- ❑ One-off community walk-throughs?
- ❑ Hopefully those just getting started contribute to projects of interest in their own right in time

Acknowledgements



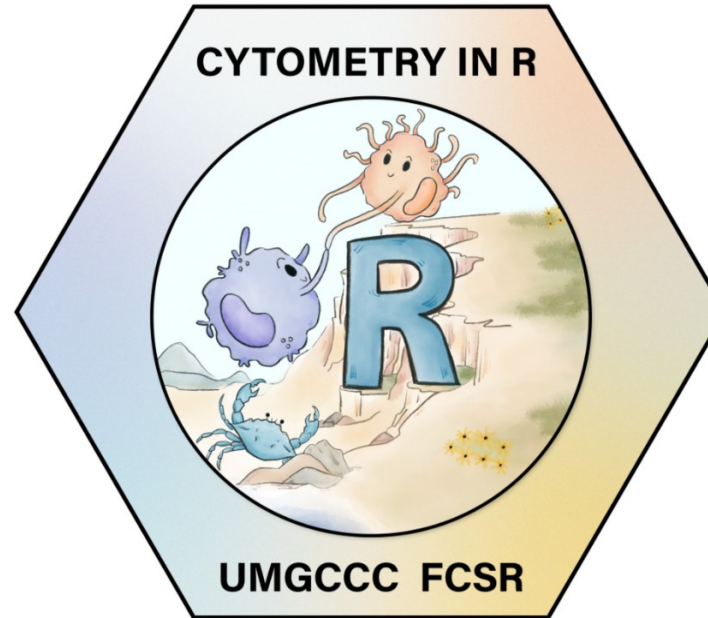
ISAC RECOGNIZED
Shared Resource Laboratory

All the open-source contributors, developers, and maintainers, that collectively made the tool sets without which this course would not be feasible

All course participants, who have had occasionally had to witness us building the plane as we fly it

Dr. Cristiana Cairo
Dr. Kirsten E. Lyke

Dr. Daniela Franco



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Cytometry
Discord



Bioconductor
Zulip

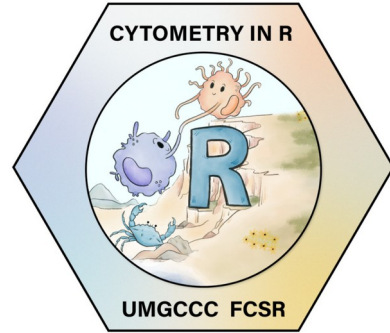
 <https://github.com/DavidRach>

 @davidrach.bsky.social

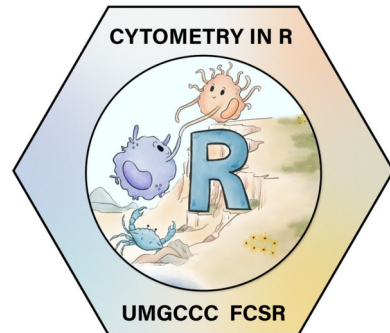


Questions?

Current Works In Progress



Website:
<https://github.com/UMGCCCFCR/CytometryInR>



YouTube @CytometryInR



Luciérnaga



Coereba